

Suspension Outboard Point (Hardpoint) Optimisation

Objective

Front upper and lower ball joint (outboard point) coordinates set the suspension's kinematic behaviour almost entirely: kingpin inclination (KPI), caster, mechanical trail, and scrub radius all fall directly out of where these two points sit in space. Those four quantities in turn drive three things that matter to the driver and to tyre performance: steering torque (driver effort), bump steer (how much the wheel steers itself as it moves through travel, independent of driver input), and dynamic camber gain (how camber angle changes with wheel travel and with steer, which sets how much contact patch is retained as the suspension moves).

This report documents the framework built to search outboard point coordinate space for a pair that improves steering torque and bump steer over the previous car (E13), for the next iteration (E14), and presents the resulting kinematic curves.

1. Approach

Outboard point coordinates cannot be chosen by solving for a target KPI, caster, trail, and scrub radius directly and independently, because all four are coupled through the same two points; moving the upper or lower ball joint to fix one metric shifts the other three simultaneously. Rather than hand-iterate this coupled system, or attempt an analytical inverse (which the four metrics do not admit cleanly, since they are nonlinear functions of the same coordinates), the problem was treated as a constrained search: generate a large pool of physically reasonable candidate coordinate pairs, simulate the resulting kinematics for each, and filter down to the candidates that satisfy the actual target metrics, rather than the four intermediate geometric quantities.

This reframes the problem from “find coordinates that produce specific KPI/caster/trail/scrub values” (a difficult inverse problem to pose correctly, since it isn't obvious in advance what combination of those four actually yields good steering torque and bump steer) to “find coordinates that produce the steering torque and bump steer behaviour we want” (a direct forward search against the metrics that actually matter).

2. Candidate Coordinate Generation

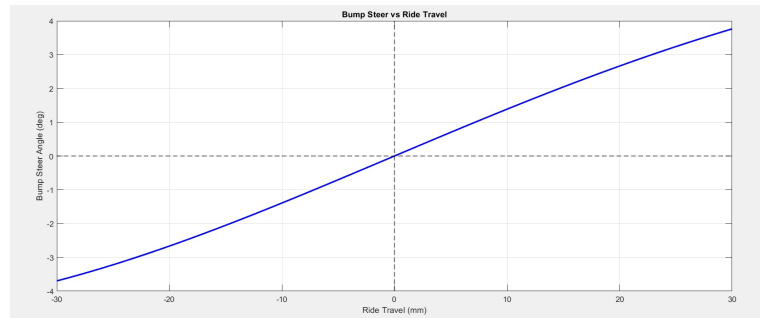
The candidate pool was built by perturbing a known-good baseline rather than sampling an unbounded space. E13's outboard point coordinates were rotated about the X and Y axes independently, in a range of $\pm 15^\circ$, in 1° steps. This produced 61 candidate coordinate sets, each a small geometric variation on a car that was already known to run and handle acceptably, keeping the search local to a

region of the design space with a reasonable prior of producing a driveable result, rather than risking convergence on a kinematically valid but practically unreasonable geometry.

3. Target Metrics

Steering torque. E13's steering torque ranged from 3.47 to 6.02 Nm (theoretical, excluding mechanical losses). Based on published reference ranges, the target for E14 was tightened to 2.5–5 Nm.

Bump steer. E13's bump steer ranged from -3.69° to 3.76° across ± 30 mm of wheel travel:



E13 bump steer

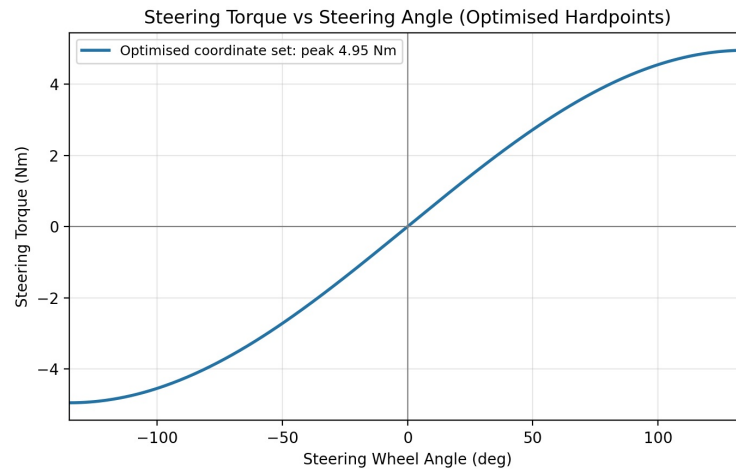
The target for E14 was to reduce this range below E13's, without a specific fixed bracket.

4. Filtering Methodology

The 61 candidates were narrowed in two sequential, hard-constraint stages rather than a single weighted score, so that no candidate could compensate for a poor result on one metric with a good result on the other:

1. **Steering torque filter.** For each candidate coordinate pair, steering torque was computed across the full steering range, and its maximum and minimum recorded. Candidates whose full torque range fell within the 2.5–5 Nm target were kept; the rest were discarded.
2. **Bump steer filter.** For the surviving candidates only, bump steer was computed across the full ride-height travel range. Candidates were kept only if their bump steer range improved on E13's baseline.

Running the bump steer filter only on candidates that had already cleared the torque constraint means the final shortlist satisfies both metrics simultaneously, rather than optimising them independently and hoping the intersection is non-empty.



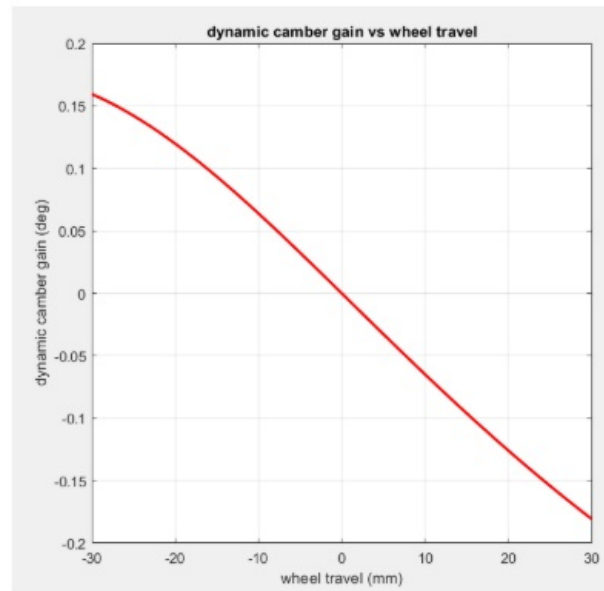
Steering torque, optimised coordinate set

This curve represents one optimised coordinate set from the shortlist, reusing the same representative curve shape used to illustrate steering torque elsewhere; it is constructed to match the achieved peak value rather than a pointwise export of raw simulation output.

5. Dynamic Camber Gain

Camber gain was evaluated separately from the two filtering metrics, characterising how much the wheel's camber angle changes both with pure vertical wheel travel (bump/droop, independent of steering) and with steering input (independent of ride height), since these are two physically distinct sources of camber change that both affect how much tyre contact patch is retained.

Camber gain vs wheel travel (bump-induced camber, at zero steer):

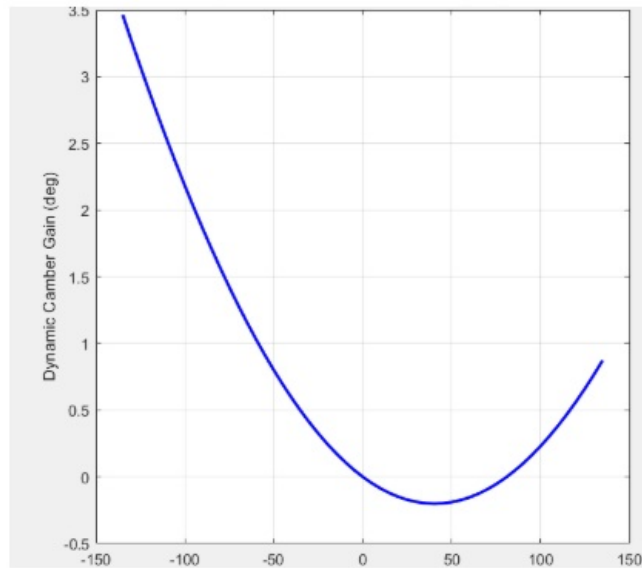


Dynamic camber gain vs wheel travel

Camber gain here is close to linear through the working range, running from roughly $+0.16^\circ$ at full droop (-30 mm) to about -0.18° at full bump (+30 mm), giving a fairly constant camber-per-millimetre gradient. This near-linearity is a desirable trait: it means the

suspension's camber compensation behaves predictably across the working travel range rather than changing rate unexpectedly near either end.

Camber gain vs steering angle (steer-induced camber, at static ride height):



Dynamic camber gain vs steering angle

This curve is not linear or symmetric. Camber gain is highest ($\sim 3.5^\circ$) at full opposite-lock, falls through a minimum of roughly -0.2° around 40° of steer, and rises again toward the opposite steering limit ($\sim 0.9^\circ$ at $+135^\circ$). The asymmetry between the two steering limits, and the fact that the minimum sits off-centre rather than at zero steer, both fall directly out of the specific outboard point geometry rather than being an artefact of the plotting; this is exactly the kind of nonlinear, non-obvious coupling that makes hand-solving for target KPI/caster/trail/scrub values impractical, and is why the coordinate search was run against directly observable output curves like this one instead.

6. Result and Current Status

The two-stage filter narrowed the 61-candidate pool to approximately 6–8 shortlisted coordinate pairs, all satisfying both the steering torque and bump steer targets simultaneously. Each of these shortlisted candidates was then evaluated on the full set of four metrics, steering torque, bump steer, bump-induced camber gain, and steer-induced camber gain, and the coordinate pair performing best across all four simultaneously was selected as the final outboard point geometry for E14.

This finalised coordinate pair has been handed off to the suspension subsystem for upright and mounting design.